

Actuarial Knowledge Test

Instructions: You have a maximum of one and a half hours (1 hour) to complete this test.

Actuarial Risk

1. Name three probability distributions that can be used to model severity and three that can be used to model frequency in an insurance portfolio. **Note: The Poisson, Logarithmic, Negative Binomial, and Pareto distributions are strictly forbidden and will not be accepted as valid answers.**

Solution

Frequency Modeling (Discrete Count Distributions)

- **Delaporte Distribution:** A powerful three-parameter mixture (Poisson-Gamma with a shifted parameter). It is recognized as the global gold standard for designing Bonus-Malus systems in automobile insurance due to its unique mathematical capability to isolate homogeneous baseline risk from the extreme overdispersion and heterogeneity of individual drivers.
- **Conway-Maxwell-Poisson (CMP) Distribution:** A sophisticated two-parameter generalization of the standard Poisson distribution. It possesses the unique property of modeling both overdispersed and underdispersed portfolios, elegantly encompassing the traditional Poisson and Geometric distributions as special cases.
- **Neyman Type A Distribution:** A compound Poisson-Poisson (or contagion) distribution. It is ideal for insurance lines where claims occur in clustered bursts, such as property or agricultural insurance, where multiple simultaneous claims stem from a single climatic event or natural disaster.

Severity Modeling (Continuous Loss Distributions)

- **Burr (Type XII) Distribution:** A highly flexible three-parameter distribution that generalizes heavy-tailed behavior. It stands as the modern industry standard in reinsurance for modeling extreme loss layers and high-severity claims.
- **Dagum Distribution:** A right-skewed, heavy-tailed continuous model adapted with great success in modern actuarial theory to fit catastrophic loss data. It outperforms standard models when estimating the exact thickness of the tail risk.
- **Log-Cauchy Distribution:** An ultra-heavy-tailed distribution with no finite moments (infinite theoretical mean and variance). It is strictly reserved for "black swan" events or extreme catastrophic risks where a single claim has the potential to destabilize the entire reserve fund.

Reserves

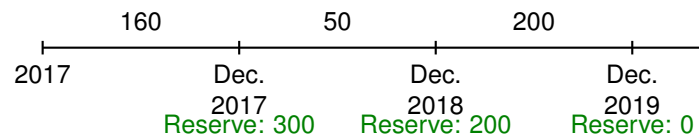
2. Consider the following claims information, stated in millions.

Claim number	Accident date	Notice date	Amount paid during			Reported claim reserve as of December 31		
			2017	2018	2019	2017	2018	2019
1	01/02/2017	22/04/2017	160	50	200	300	200	0
2	01/10/2017	15/12/2017	40	50	200	100	100	0
3	01/05/2018	23/07/2018		200	100		100	100
4	15/12/2018	29/12/2018		100	300		500	200
5	01/04/2019	15/06/2019			100			100

- a) Construct the triangle of cumulative incurred claim amounts by accident year.

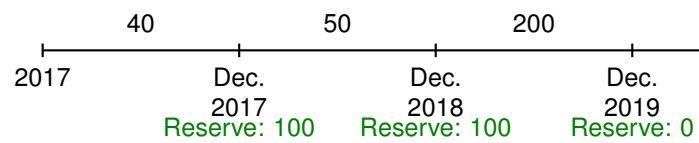
Solution

Claim 1



Valuation year	Cumulative incurred	Calculation
2017	460	$160 + 300$
2018	410	$160 + 50 + 200$
2019	410	$160 + 50 + 200 + 0$

Claim 2



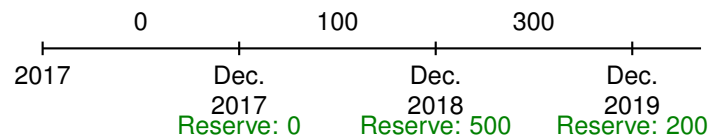
Valuation year	Cumulative incurred	Calculation
2017	140	$40 + 100$
2018	190	$40 + 50 + 100$
2019	290	$40 + 50 + 200 + 0$

Claim 3



Valuation year	Cumulative incurred	Calculation
2017	0	$0 + 0$
2018	300	$0 + 200 + 100$
2019	400	$0 + 200 + 100 + 100$

Claim 4



Valuation year	Cumulative incurred	Calculation
2017	0	$0 + 0$
2018	600	$0 + 100 + 500$
2019	600	$0 + 100 + 300 + 200$

Cumulative incurred claims triangle

	0	1	2
2017	600	600	700
2018	900	1000	
2019	200		

- b) Determine the IBNR using the Chain-Ladder method, and justify each step.

Solution

1. Development factors

The observed individual age-to-age factors are

$$\frac{600}{600} = 1, \quad \frac{1000}{900} = 1.111111, \quad \frac{700}{600} = 1.166667.$$

The selected Chain-Ladder factors are calculated as volume-weighted averages:

$$f_{0 \rightarrow 1} = \frac{600 + 1000}{600 + 900} = 1.066667, \quad f_{1 \rightarrow 2} = \frac{700}{600} = 1.166667.$$

	0-1	1-2
Simple average	1.055556	1.166667
Selected LDF	1.066667	1.166667

2. Completion of the cumulative incurred triangle

The missing cumulative amounts are projected recursively using $\hat{C}_{i,j+1} = C_{i,j} f_{j \rightarrow j+1}$:

$$\hat{C}_{2018,2} = 1000(1.166667) = 1166.667,$$

$$\hat{C}_{2019,1} = 200(1.066667) = 213.3333,$$

$$\hat{C}_{2019,2} = 213.3333(1.166667) = 248.8889.$$

	0	1	2
2017	600	600	700
2018	900	1000	1166.667
2019	200	213.3333	248.8889

3. Ultimate claims and IBNR

For each accident year, the ultimate claim is the amount in the final development column. The IBNR is the difference between that ultimate and the latest reported cumulative incurred amount on the observed diagonal.

Accident year	Ultimate	Reported incurred	IBNR
2017	700.0000	700.0000	0.0000
2018	1166.6667	1000.0000	166.6667
2019	248.8889	200.0000	48.8889
Total			215.5556

Therefore, the estimated total IBNR is

$\text{IBNR} = 215.5556 \text{ million.}$

3. List some of the technical reserves that insurance companies must manage under Decree 2973.

Solution

The six technical reserves that insurance companies must manage under Decree 2973 are:

1. **Unearned Premium Reserve (Reserva de Riesgos en Curso):** Established to fulfill future obligations arising from active policies. It consists of:
 - *Unearned premium reserve:* The portion of the written premium corresponding to the unexpired risk period.
 - *Premium deficiency reserve:* A supplement used when the unearned premium is insufficient to cover future risks and expenses.
2. **Mathematical Reserve:** Set aside to cover commitments under individual life insurance and annuity policies calculated using level premiums.
3. **Asset Adequacy Reserve:** Created to offset any shortfall when expected asset cash flows fail to match the expected liability cash flows of the mathematical reserve.
4. **Outstanding Claims Reserve:** Allocated to pay claims that have already occurred. It is divided into:
 - *Reported claims reserve:* Funds for claims already reported and their associated loss-adjustment expenses.
 - *Incurred But Not Reported (IBNR) reserve:* An estimate for claims that have already occurred but have not yet been reported to the insurer.
5. **Claims Fluctuation Reserve:** Designed to mitigate wide fluctuations and statistical deviations from expected claim experience.
6. **Catastrophic Risk Reserve:** Established to cover risks arising from catastrophic events characterized by low frequency and high severity.

Life Insurance

4. Consider the following extract from the RV08 mortality table for healthy annuitants.

Age	Men				Women			
	l_x	d_x	q_x	p_x	l_y	d_y	q_y	p_y
...	...	...	...	...	...	...	...	...
35	986,216	1,102	0.001117	0.99888	992,247	622	0.000627	0.99937
36	985,114	1,172	0.001190	0.99881	991,625	662	0.000668	0.99933
37	983,942	1,249	0.001269	0.99873	990,963	705	0.000711	0.99929
38	982,693	1,333	0.001356	0.99864	990,258	753	0.000760	0.99924
39	981,360	1,424	0.001451	0.99855	989,505	806	0.000815	0.99919
40	979,936	1,525	0.001556	0.99844	988,699	863	0.000873	0.99913
...	...	...	...	...	...	...	...	...

- a) Determine the probability that a man currently aged 35 dies before reaching his 38th birthday, and the probability that a woman currently aged 35 survives at least to age 39.

Solution

For the man aged 35, the probability of dying before reaching age 38 is

$${}_3q_{35} = 1 - {}_3p_{35} = 1 - \frac{l_{38}}{l_{35}} = 1 - \frac{982,693}{986,216} = 0.003572239753.$$

Thus, the probability is approximately 0.3572%.

For the woman aged 35, the probability of surviving at least to age 39 is

$${}_4p_{35} = \frac{l_{39}}{l_{35}} = \frac{989,505}{992,247} = 0.9972365752.$$

Thus, the probability is approximately 99.7237%.

- b) Calculate the present value of a two-year term life insurance policy for a man currently aged 37. The policy pays \$100,000,000 at the end of the policy year in which the insured dies. Use an interest rate of 10%.

Solution

For a two-year term insurance with benefit $S = 100,000,000$ and annual effective interest rate $i = 10\%$, the discount factor is

$$v = \frac{1}{1+i} = \frac{1}{1.10}.$$

The benefit is paid at the end of the year of death. Therefore, its actuarial present value is

$$A_{37:\overline{2}|}^1 = vq_{37} + v^2p_{37}q_{38}.$$

The first term corresponds to death between ages 37 and 38, and the second corresponds to survival to age 38 followed by death between ages 38 and 39.

Using $q_{37} = 0.001269$, $q_{38} = 0.001356$, and

$$p_{37} = \frac{l_{38}}{l_{37}} = \frac{982,693}{983,942},$$

the expected present value is

$$\begin{aligned} \text{EPV} &= 100,000,000 \left[\frac{0.001269}{1.10} + \frac{\left(\frac{982,693}{983,942} \right) (0.001356)}{(1.10)^2} \right] \\ &= \boxed{227,287.4972}. \end{aligned}$$

Programming

You have two databases containing information about a personal accident insurance product (covering medical expenses and transportation to healthcare facilities for potential victims of traffic accidents). These databases are named POLIZAS and SINIESTROS and are described below:

POLIZAS: Contains consolidated historical information on issued policies and includes the following variables

IDPOLIZA = Number identifying the policy

FECEXP = Policy issue date

FECIVIG = Policy effective date

FECVEN = Policy expiration date

IDTOMADOR = Number identifying the policyholder

PRIMA = Written premium amount

TIPORIESGO = Risk segment to which the policy belongs (only two values: "motorcycles" and "non-motorcycles")

SINIESTROS: Contains information on payments made under the two coverages described above and includes the following variables

IDPOLIZA = Number identifying the policy
 IDACCIDENTE = Number identifying the accident
 IDVICTIMA = Number identifying the victim
 FCHOCUR = Accident occurrence date
 FCHAVI = Accident notice date
 FCHPAGO = Payment date
 COBERTURA = Affected coverage (only two values: GM (medical expenses) and TR (transportation))
 VLRPAGO = Amount paid (may contain negative values, which must be interpreted as payment reversals)

5. Assume that these databases have been loaded into R, Python, or SQL and have no data-quality issues. Write the **programming code** in one of these languages that performs each of the following tasks:

- a) Determine the number of policies issued in 2020 by risk type.
- b) Determine the number of policies in force on March 25, 2020.
- c) Assign the corresponding risk type of the affected policy to each record in the SINIESTROS database and save the result in a table named SINIESTROS2 (which must contain exactly one additional column relative to SINIESTROS: the risk type).
- d) Determine the total amount paid per victim and create a table named PAGXVIC with the following structure:

IDVICTIMA	GM	TR

GM must contain the total amount paid per victim under the medical-expense coverage, and TR must contain the total amount paid per victim under the transportation coverage.

Solution

The R script with the complete solution for this section can be found in the project folder.

Machine Learning

6. Define the following concepts related to generalized linear models:

- Random Component:
- Systematic Component:
- Link Function:
- Logistic Regression:
- Deviance:
- Overdispersion:
- AIC – BIC:

Solution

Random Component: This component identifies the response or target variable (Y) and specifies its probability distribution assumption. Within the framework of Generalized Linear Models (GLMs), this distribution must strictly belong to the classical exponential family (such as Normal, Binomial, Poisson, Gamma, or Tweedie), capturing the intrinsic randomness and uncertainty of the underlying process.

Systematic Component: This is the deterministic structure of the model that aggregates the explanatory variables or specific predictors ($X_1, X_2, \dots, X_k$). It is formally expressed as the linear predictor (denoted by the Greek letter η [eta]), which linearly combines the effects of the covariates with their corresponding regression coefficients (β):

$$\eta = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k = X\beta$$

Link Function: A link function is a monotonic and differentiable mathematical function, denoted as $g(\cdot)$, that analytically connects the expected value of the random component ($\mu = E[Y]$) to the linear predictor of the systematic component (η), formalized as $g(\mu) = \eta$. Its primary purpose is to map the domain of the mean parameter space onto the linear scale, ensuring that the model does not produce predictions outside logical boundaries (such as negative frequencies or claim costs).

Logistic Regression: This is a special case of Generalized Linear Models designed specifically for modeling binary or dichotomous response variables ($Y \in \{0, 1\}$). It assumes a random component with a Binomial distribution and utilizes the logit function as its canonical link function, which models the natural logarithm of the odds of the event occurring:

$$g(p) = \ln\left(\frac{p}{1-p}\right) = X\beta$$

Deviance: This is a statistical metric used to evaluate the overall goodness-of-fit of a GLM based on the maximum likelihood principle. It quantifies the discrepancy or "lack of fit" between the proposed model and the saturated model (a theoretical model that perfectly replicates every single observed data point). Mathematically, it is defined as:

$$D = 2(\ell(\text{Saturated Model}) - \ell(\text{Proposed Model}))$$

Where ℓ represents the log-likelihood. A lower deviance indicates a better fit of the model to the data.

Overdispersion: This is a statistical phenomenon that occurs when the observed variance in the sample data is significantly greater than the theoretical variance parameterized by the chosen probability distribution. It is highly common when fitting models under the Poisson distribution, where equidispersion ($E[Y] = \text{Var}(Y)$) is strictly assumed. Ignoring overdispersion leads to underestimated standard errors of the coefficients, which invalidates hypothesis testing.

AIC BIC (Information Criteria): Selection criteria that balance goodness-of-fit and model complexity, defined as:

$$\begin{aligned} AIC &= -2\ell + 2k \\ BIC &= -2\ell + k \ln n \end{aligned}$$

Where ℓ is the log-likelihood, k is the number of parameters, and n is the sample size. Among models fitted to the same dataset, the model with the lowest value is preferred. BIC penalizes model complexity more heavily than AIC when n is large.

7. Name some goodness-of-fit tests that you would use to assess the distribution of a variable of interest.

Solution

Shapiro-Wilk Test: Best for normality. Highly powerful for small sample sizes ($n < 50$).

Kolmogorov-Smirnov (K-S) Test: Compares empirical and theoretical CDFs. Sensitive to shape, scale, and location. Best when parameters are known.

Lilliefors Test: Modified K-S test. Use this when distribution parameters (mean, variance) are estimated from the data.

Anderson-Darling Test: Modification of the K-S test. Gives more weight to the tails of the distribution.

Chi-Square Goodness-of-Fit Test: Compares observed frequencies against expected frequencies. Requires minimum expected counts per bin (usually ≥ 5)

8. Assume a sample of 80,994 policyholders with automobile insurance policies from Company X. The database contains the following information:

Abbreviation	Variable	Categories
SIN2	Number of Claims	
V9	Sex	woman; 0 = man (dichotomous)
V10	Driving area	urban; 0 = otherwise
V13	Medium experience	4–14 years; 0 = otherwise
V14	Maximum experience	more than 15 years; 0 = otherwise
V17	Age	30 or older; 0 = otherwise
V18	Additional coverages	yes; 0 = no
V19	Comprehensive coverage	yes; 0 = no
V20	Engine power	more than 120 hp; 0 = no

The following R code is used:

```
Pois <- glm(data$SIN2 ~ data$V9 + data$V10 + data$V13 + data$V14 +  
            data$V17 + data$V18 + data$V19 + data$V20,  
            family = poisson(link="log"))
```

Based on this code, describe:

- a) The model used (formula).

Solution

The code specifies a Poisson Regression Model (a type of Generalized Linear Model) used for count data. The mathematical formula for the model expresses the log of the expected number of claims as a linear combination of the predictor variables:

$$\ln(\lambda_i) = \beta_0 + \beta_1(V9_i) + \beta_2(V10_i) + \beta_3(V13_i) + \beta_4(V14_i) + \beta_5(V17_i) + \beta_6(V18_i) + \beta_7(V19_i) + \beta_8(V20_i)$$

- b) The link function

Solution

The link function used is the logarithmic link function ($\text{link} = \text{"log"}$):

$$g(\mu) = \ln(\mu)$$

Purpose: It connects the linear predictor ($\beta_0 + \beta_1 X_1 + \dots$) to the expected mean (μ) of the Poisson distribution.

Functionality: Since count data (number of claims) must strictly be non-negative ($\mu > 0$), taking the exponent of the linear combination ensures that the predicted values are always positive, regardless of the values of the coefficients or independent variables.

The results of the fitted model are presented below:

```

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.47468  0.16552 -14.951 < 2e-16 ***
data$V9      -0.03090  0.06481  -0.477  0.633553
data$V10     -0.04861  0.04924  -0.987  0.323528
data$V13     -0.22010  0.14129  -1.558  0.119279
data$V14     -0.35123  0.15267  -2.301  0.021413 *
data$V17      0.11895  0.09924   1.199  0.230666
data$V18      0.23387  0.06572   3.559  0.000373 ***
data$V19      0.07946  0.05342   1.488  0.136840
data$V20      0.16924  0.06734   2.513  0.011961 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 9781.9 on 24207 degrees of freedom
Residual deviance: 9737.3 on 24199 degrees of freedom
AIC: 13174

```

c) What can you infer from this model?

Impact of Significant Risk Factors:

- Experience (V14): Drivers with maximum experience (more than 15 years) show a significant reduction in the expected number of claims compared to the reference group.
- Coverage and Engine Power (V18 and V20): Having additional coverages and driving a vehicle with engine power exceeding 120 hp are both associated with a significant increase in the expected number of claims.

Variables with No Proven Impact: Factors such as gender (V9), driving area (V10), medium experience (V13), age (V17), and comprehensive coverage (V19) do not show a statistically relevant effect on the number of claims within this sample.

Low Global Explanatory Power: The difference between the Null deviance (9781.9) and the Residual deviance (9737.3) is extremely small (a reduction of only 44.6 units). This indicates that while specific individual variables are meaningful, the current set of predictors explains a very small fraction of the overall variance in insurance claims.

d) Which variables are statistically significant?

Assuming a standard significance level of $\alpha = 0.05$ (based on the $\text{Pr}(>|z|)$ column and the significance stars code):

- $\text{dataV18}(p\text{-value} = 0.000373)$ - *Highly significant* (***)
- $\text{dataV20}(p\text{-value} = 0.011961)$ - *Statistically significant* (*)
- $\text{dataV14}(p\text{-value} = 0.021413)$ - *Statistically significant* (*)

(Note: The Intercept is also highly significant ($p\text{-value} < 2 \times 10^{-16}$), but it represents the baseline constant of the model rather than an explanatory variable).

e) What actions would you take to improve the model?

Solution

- **Data Quality and Sample Discrepancy:** Investigate why the dataset dropped from 80,994 to 24,208 records. Verify whether the approximately 70% loss of observations was caused by R's handling of missing values and assess whether it introduces severe selection bias.
- **Exposure and Offset Integration:** Incorporate an exposure measure as an offset term,

```
+ offset(log(exposure_time)),
```

so that the model estimates claim frequency per vehicle-year rather than modeling raw claim counts.

- **Dispersion and Zero-Inflation Diagnostics:** Test for overdispersion or underdispersion and examine the proportion of zero-claim observations. If necessary, compare alternatives such as Quasi-Poisson, Negative Binomial, or zero-inflated models.
- **Actuarial-Driven Variable Selection:** Group sparse categories and select predictors using actuarial judgment and risk theory, rather than relying exclusively on individual p-values.
- **Validation and Pricing Stability:** Implement out-of-sample validation, such as a train–test split or cross-validation; compare candidate models using AIC and BIC when applicable; and assess temporal stability to ensure that predictions are structurally sound for insurance ratemaking.